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COURSE CODE: SBT-DF204
LAB NAME: CASE STUDY 3 INDIVIDUAL MEMORY FORENSICS INVESTIGATION
DATE: 19/10/2026
INSTRUCTOR: AMINU IDRIS

This individual case study asks you to examine a Windows 7 memory image and reconstruct an evidence-based account of a suspected illegal file transfer. Use Volatility 2 to identify relevant processes, network artefacts, command history, removable-media records, user-profile evidence, and File Explorer traces. Your report must clearly separate direct observations from inferences about a transfer or the person who used the workstation.

Case scenario

Investigators obtained a Windows 10 memory image from a workstation after a confidential document may have been transferred without authorization. The inquiry is whether browser activity, a command prompt, a removable device, and a user profile form a defensible sequence. This is an offline examination of the supplied material. It does not authorize contact with any address, host, account, file-sharing service, or person that may appear in the image.

Case materials

The supplied case file is a Windows 7 memory image named memdumpWin7.mem. The teaching deck demonstrates Volatility 2 methods. In the ICDFA Kali lab, start Volatility 2 with python2 vol.py. Create a working copy before analysis. Record the source, original file name, file size, SHA-256 value, and tool version. Select and justify the memory profile from the actual image output instead of assuming one from a screenshot.

What you must submit

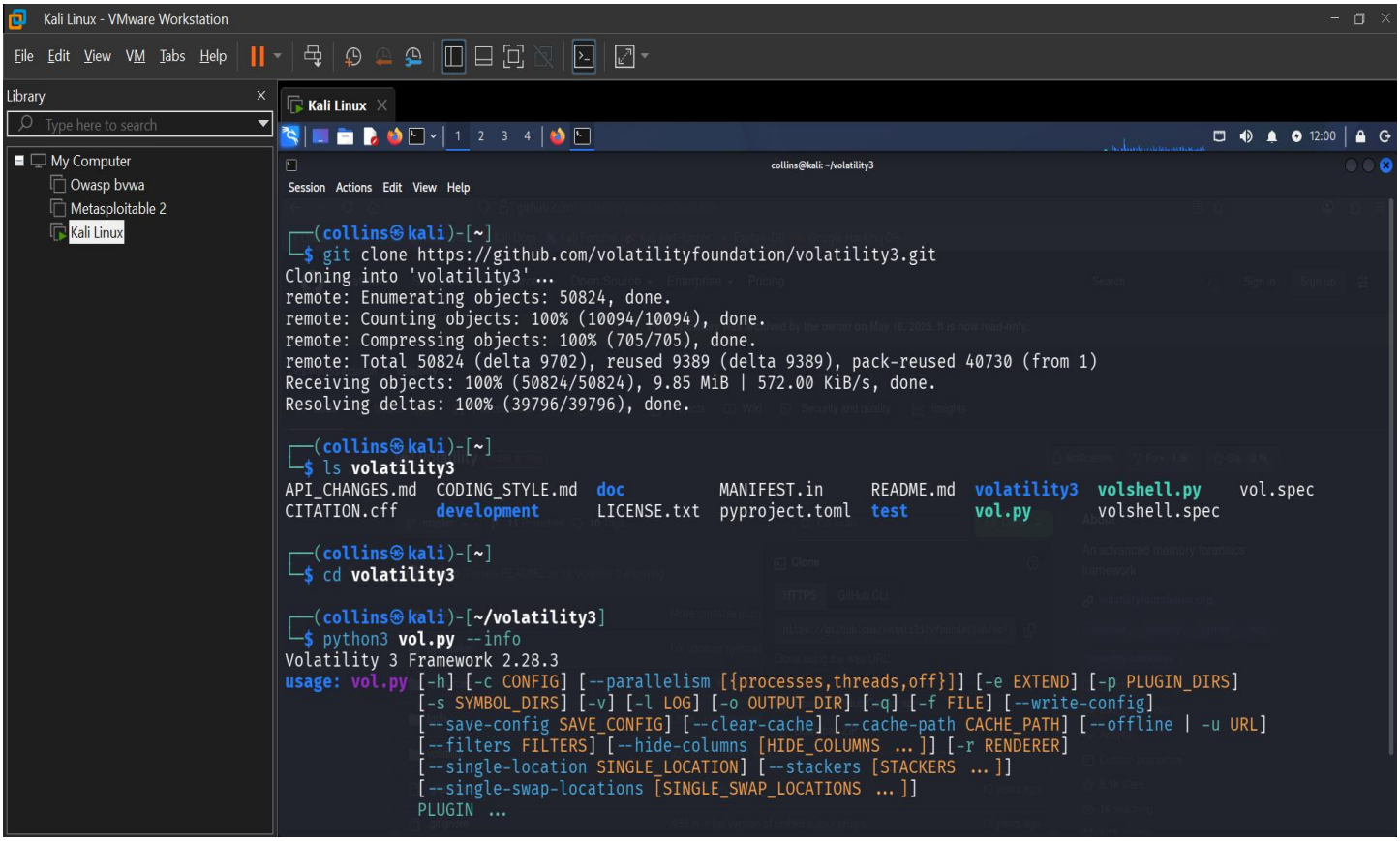
Item	Required output	What to provide
1	Forensic working copy	Acquire the supplied memory image. Record its source, original file name, byte size, SHA-256 value, and the location of your read-only original and working copy.
2	Volatility evidence set	Preserve output from profile identification, process and network analysis, command history, USB registry and MountPoints2 checks, and ShellBag analysis. Label every output with the plugin, profile, and command used.
3	Reconstructed timeline	Build a time-ordered account of the suspected transfer. Cite process IDs, registry paths, offsets or record details, timestamps, and evidence-log IDs where relevant.
4	Student report	Submit one report in PDF or DOCX format with labelled screenshots, a completed evidence log, and a timeline appendix.

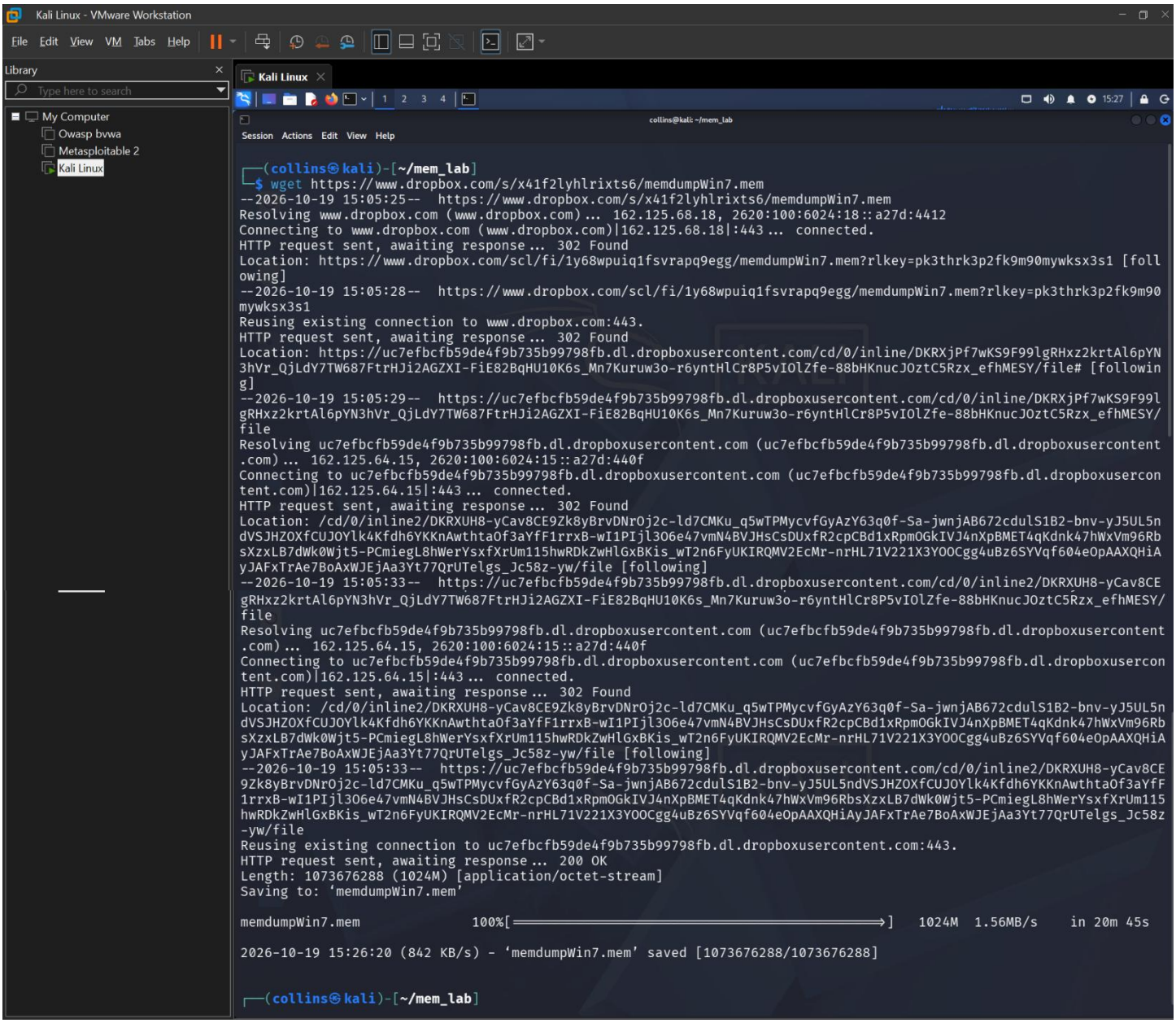
Investigation task

Examine the memory image methodically. Base every finding on identifiable Volatility output, including the plugin, profile, command, timestamp, record, registry path, process ID, or offset that supports it. A recovered string, process

name, command, or registry key can be important, but it may not prove that a transfer completed or that a particular person performed it. Use the questions below to structure your report.

No	Investigation question	Minimum evidence to discuss
1	What evidence image did you acquire and how did you preserve it?	Record the source, file name, file size, SHA-256 value, working-copy method, file-access precautions, and tools used.
2	What operating-system profile and image time are supported?	Provide imageinfo output, the candidate and selected profile, the image time, and the time zone. Explain how you validated the profile.
3	What user and account artefacts are relevant?	Identify the account or profile records examined, their SIDs or registry-hive paths, and the limit between profile evidence and proof of a person at the keyboard.
4	What process and network evidence relates to a potential download or transfer?	Identify relevant process IDs, process names, local and foreign endpoints, connection state, and memory strings or output records. Do not contact any endpoint.
5	What does command history show?	Use cmdscan or consoles output to identify relevant commands, the command process, path or drive references, and the timing or correlation evidence. Explain any limit on proving success.
6	What removable-media evidence is present?	Use USBSTOR, DeviceClasses, MountedDevices, and MountPoints2 evidence as available to identify device, drive-letter, and user-profile relationships. State what each artefact does and does not establish.
7	What File Explorer evidence supports the sequence?	Use ShellBags to identify material folder paths and recorded access or preference times. Distinguish folder-view evidence from a file-modification timestamp.
8	What conclusion can you defend?	Create a UTC timeline and state a cautious conclusion with a confidence level and at least three limitations or alternative explanations.





Volatility 2 examination starting points

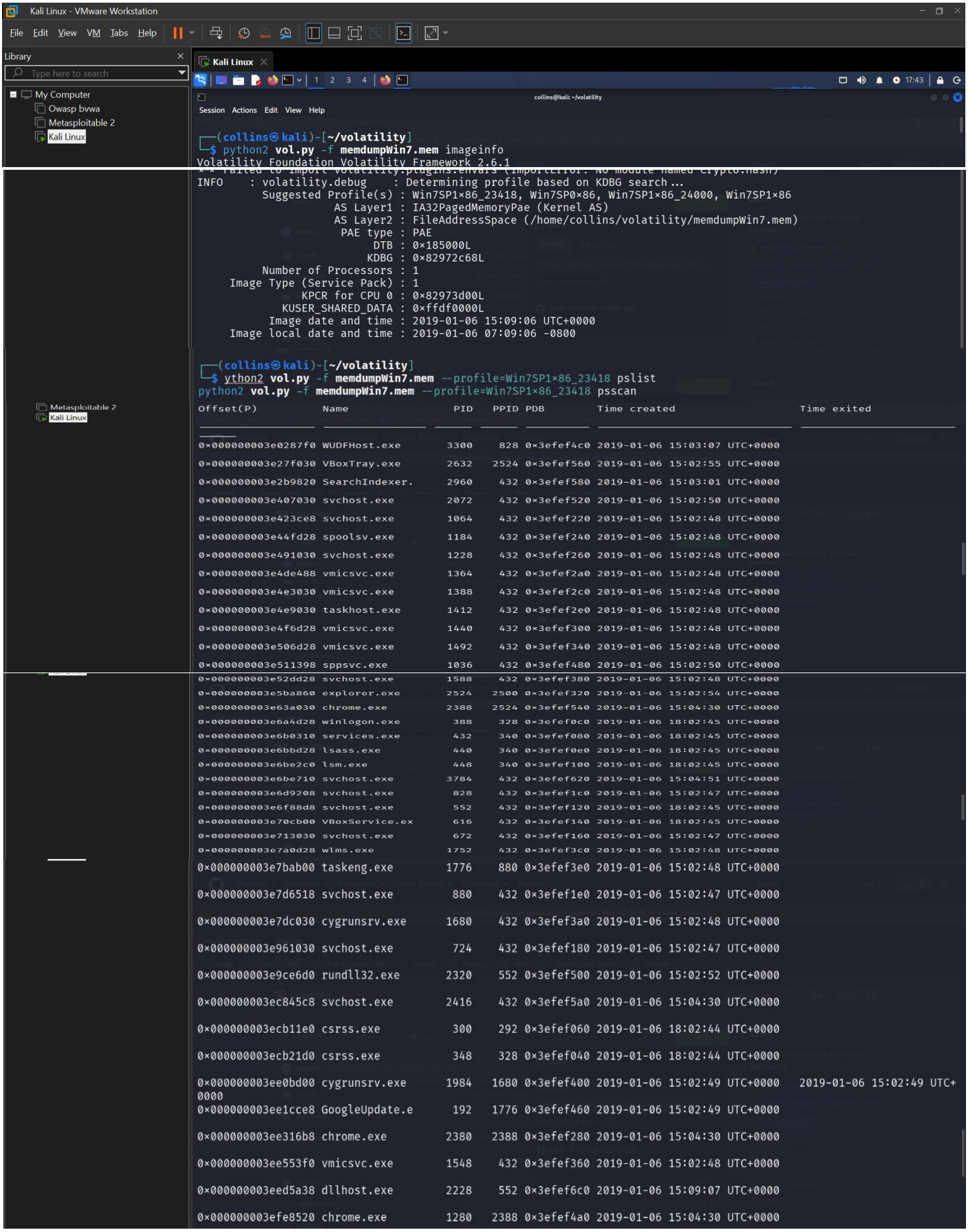
Use the commands below as starting points. Replace <profile> with the profile you select from imageinfo and record any variation needed for the authorised lab environment. The commands guide your examination; the returned records provide the evidence.

Purpose	Starting command
Identify the image profile and capture time	`python2 vol.py -f memdumpWin7.mem imageinfo`
List active and residual processes	`python2 vol.py -f memdumpWin7.mem --profile=<profile> pslist` `python2 vol.py -f memdumpWin7.mem --profile=<profile> psscan`
Review network artefacts	`python2 vol.py -f memdumpWin7.mem --profile=<profile> netscan`
Review command history	`python2 vol.py -f memdumpWin7.mem --profile=<profile> cmdscan` `python2 vol.py -f memdumpWin7.mem --profile=<profile> consoles`

Locate USB and drive-mapping artefacts	`python2 vol.py -f memdumpWin7.mem --profile=<profile> hivelist`
	`python2 vol.py -f memdumpWin7.mem --profile=<profile> printkey -K "ControlSet001\Enum\USBSTOR"`
	`python2 vol.py -f memdumpWin7.mem --profile=<profile> printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"`

Review File Explorer history	`python2 vol.py -f memdumpWin7.mem --profile=<profile> shellbags --output=html --output-file=shellbags.html`

ANSWER:


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bvwa
Metasploitable 2
Kali Linux
Kali Linux
Session Actions Edit View Help
collins@kali: ~/volatility
(collins㉿kali)-[~/volatility]
$ python2 vol.py -f memdumpWin7.mem imageinfo
Volatility Foundation Volatility Framework 2.6.1
INFO : volatility.debug : Determining profile based on KDBG search...
Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
AS Layer1 : IA32PagedMemoryPae (Kernel AS)
AS Layer2 : FileAddressSpace (/home/collins/volatility/memdumpWin7.mem)
PAE type : PAE
DTB : 0x185000L
KDBG : 0x82972c68L
Number of Processors : 1
Image Type (Service Pack) : 1
KPCR for CPU 0 : 0x82973d00L
KUSER_SHARED_DATA : 0xffdf0000L
Image date and time : 2019-01-06 15:09:06 UTC+0000
Image local date and time : 2019-01-06 07:09:06 -0800
(collins㉿kali)-[~/volatility]
$ ython2 vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 pslist
python2 vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 psscan
Offset(P) Name PID PPID PDB Time created Time exited
0x000000003e0287f0 WUDFHost.exe 3300 828 0x3efef4c0 2019-01-06 15:03:07 UTC+0000
0x000000003e27f030 VBoxTray.exe 2632 2524 0x3efef560 2019-01-06 15:02:55 UTC+0000
0x000000003e2b9820 SearchIndexer. 2960 432 0x3efef580 2019-01-06 15:03:01 UTC+0000
0x000000003e407030 svchost.exe 2072 432 0x3efef520 2019-01-06 15:02:50 UTC+0000
0x000000003e423ce8 svchost.exe 1064 432 0x3efef220 2019-01-06 15:02:48 UTC+0000
0x000000003e44fd28 spoolsv.exe 1184 432 0x3efef240 2019-01-06 15:02:48 UTC+0000
0x000000003e491030 svchost.exe 1228 432 0x3efef260 2019-01-06 15:02:48 UTC+0000
0x000000003e4de488 vmicsvc.exe 1364 432 0x3efef2a0 2019-01-06 15:02:48 UTC+0000
0x000000003e4e3030 vmicsvc.exe 1388 432 0x3efef2c0 2019-01-06 15:02:48 UTC+0000
0x000000003e4e9030 taskhost.exe 1412 432 0x3efef2e0 2019-01-06 15:02:48 UTC+0000
0x000000003e4f6d28 vmicsvc.exe 1440 432 0x3efef300 2019-01-06 15:02:48 UTC+0000
0x000000003e506d28 vmicsvc.exe 1492 432 0x3efef340 2019-01-06 15:02:48 UTC+0000
0x000000003e511398 sppsvc.exe 1036 432 0x3efef480 2019-01-06 15:02:50 UTC+0000
0x000000003e52dd28 svchost.exe 1588 432 0x3efef380 2019-01-06 15:02:48 UTC+0000
0x000000003e5ba860 explorer.exe 2524 2500 0x3efef320 2019-01-06 15:02:54 UTC+0000
0x000000003e63a030 chrome.exe 2388 2524 0x3efef540 2019-01-06 15:04:30 UTC+0000
0x000000003e6a4d28 winlogon.exe 388 328 0x3efef0c0 2019-01-06 18:02:45 UTC+0000
0x000000003e6b0310 services.exe 432 340 0x3efef080 2019-01-06 18:02:45 UTC+0000
0x000000003e6bbd28 lsass.exe 440 340 0x3efef0e0 2019-01-06 18:02:45 UTC+0000
0x000000003e6be2c0 lsm.exe 448 340 0x3efef100 2019-01-06 18:02:45 UTC+0000
0x000000003e6be710 svchost.exe 3784 432 0x3efef620 2019-01-06 15:04:51 UTC+0000
0x000000003e6d9208 svchost.exe 828 432 0x3efef1c0 2019-01-06 15:02:47 UTC+0000
0x000000003e6f88d8 svchost.exe 552 432 0x3efef120 2019-01-06 18:02:45 UTC+0000
0x000000003e70cb00 VBoxService.ex 616 432 0x3efef140 2019-01-06 18:02:45 UTC+0000
0x000000003e713030 svchost.exe 672 432 0x3efef160 2019-01-06 15:02:47 UTC+0000
0x000000003e7a0d28 wlms.exe 1752 432 0x3efef3c0 2019-01-06 15:02:48 UTC+0000
0x000000003e7bab00 taskeng.exe 1776 880 0x3efef3e0 2019-01-06 15:02:48 UTC+0000
0x000000003e7d6518 svchost.exe 880 432 0x3efef1e0 2019-01-06 15:02:47 UTC+0000
0x000000003e7dc030 cygrunsrv.exe 1680 432 0x3efef3a0 2019-01-06 15:02:48 UTC+0000
0x000000003e961030 svchost.exe 724 432 0x3efef180 2019-01-06 15:02:47 UTC+0000
0x000000003e9ce6d0 rundll32.exe 2320 552 0x3efef500 2019-01-06 15:02:52 UTC+0000
0x000000003ec845c8 svchost.exe 2416 432 0x3efef5a0 2019-01-06 15:04:30 UTC+0000
0x000000003ecb11e0 csrss.exe 300 292 0x3efef060 2019-01-06 18:02:44 UTC+0000
0x000000003ecb21d0 csrss.exe 348 328 0x3efef040 2019-01-06 18:02:44 UTC+0000
0x000000003ee0bd00 cygrunsrv.exe 1984 1680 0x3efef400 2019-01-06 15:02:49 UTC+0000 2019-01-06 15:02:49 UTC+0000
0x000000003ee1cce8 GoogleUpdate.e 192 1776 0x3efef460 2019-01-06 15:02:49 UTC+0000
0x000000003ee316b8 chrome.exe 2380 2388 0x3efef280 2019-01-06 15:04:30 UTC+0000
0x000000003ee553f0 vmicsvc.exe 1548 432 0x3efef360 2019-01-06 15:02:48 UTC+0000
0x000000003eed5a38 dllhost.exe 2228 552 0x3efef6c0 2019-01-06 15:09:07 UTC+0000
0x000000003efe8520 chrome.exe 1280 2388 0x3efef4a0 2019-01-06 15:04:30 UTC+0000

```
┌──(collins㉿kali)-[~/volatility]
└─$ python2 vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 cmdscan
python2 vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 consoles
Volatility Foundation Volatility Framework 2.6.1
**************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
----
Screen 0x3360a0 X:80 Y:300
Dump:

**************************************************
ConsoleProcess: conhost.exe Pid: 3988
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
----
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
----
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 at 0x10dc40: ipconfig
Cmd #1 at 0x107f90: cd Downloads
Cmd #2 at 0x1147a0: copy secret_file.docx F:
----
Screen 0xf63b8 X:80 Y:300
Dump:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1571:7b34:b2b1:bbe4%18
   IPv4 Address. . . . . . . . . . . : 192.168.56.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Tunnel adapter isatap.{7FD51246-C28D-46B6-AA98-EB2C150C388B}:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Tunnel adapter Teredo Tunneling Pseudo-Interface:

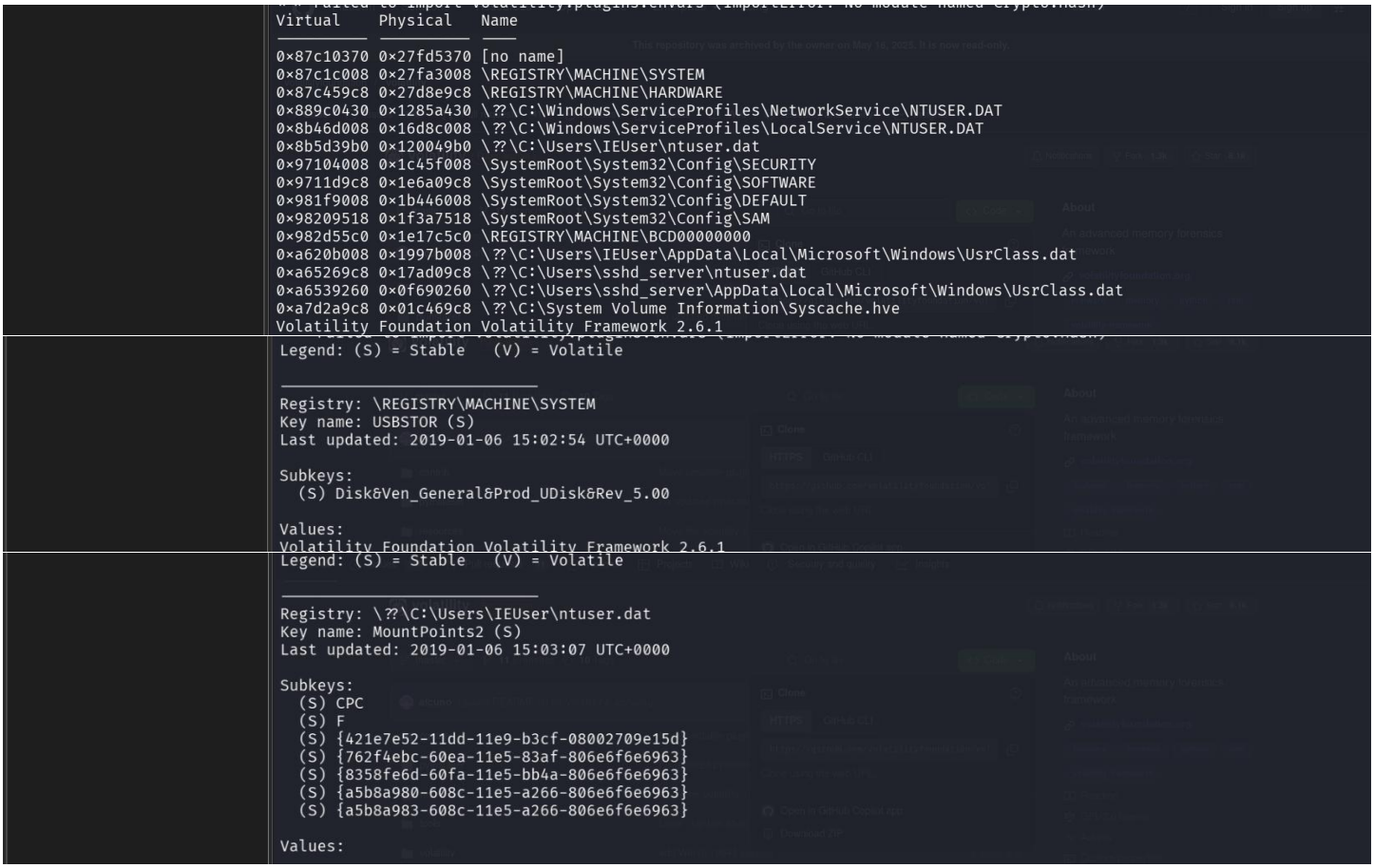
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\IEUser>cd Downloads

C:\Users\IEUser\Downloads>copy secret_file.docx F:
        1 file(s) copied.

C:\Users\IEUser\Downloads>

┌──(collins㉿kali)-[~/volatility]
└─$ python2 vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist
python2 vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Enum\USBSTOR"
python2 vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "Software\Microsoft\Windows\CurrentVersion\
Explorer\MountPoints2"
Volatility Foundation Volatility Framework 2.6.1
```



```
Virtual    Physical   Name
---------- ---------- ----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve
Volatility Foundation Volatility Framework 2.6.1
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) Disk&Ven_General&Prod_UDisk&Rev_5.00

Values:
Volatility Foundation Volatility Framework 2.6.1
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
  (S) CPC
  (S) F
  (S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
  (S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
  (S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
  (S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
  (S) {a5b8a983-608c-11e5-a266-806e6f6e6963}

Values:
```

Evidence interpretation safeguards

- Validate a profile suggested by imageinfo with usable output from multiple relevant plugins and record the profile you selected.
- Treat network endpoints and process ownership as artefacts to corroborate. A closed connection or an unidentified PID needs careful interpretation.
- Assess a command-history entry with its process, timing, and related artefacts. A command string alone may not demonstrate that the copy operation completed.
- Use USBSTOR, MountedDevices, DeviceClasses, and MountPoints2 records to explain device, drive, and profile relationships. These records alone may not prove physical possession or a user identity at a particular time.
- Treat ShellBag times as folder-view or preference evidence unless other artefacts establish a different event.

Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and case title.
2. Executive conclusion. State the supported conclusion in a short, cautious paragraph and include a confidence level.
3. Evidence acquisition and integrity. Record the memory-image source, file name, size, hash, working-copy method, and tools used.
4. Method and profile selection. Explain the imageinfo result, selected profile, Volatility version, plugin commands, and time-zone treatment.

5. Findings. Answer each investigation question with Volatility output, registry paths, process IDs, offsets or records, timestamps, and evidence-log IDs.
6. Timeline. Present the material sequence in UTC and distinguish each observed artefact from your interpretation of its meaning.
7. Attribution assessment. Explain limitations, including memory-capture timing, profile versus human identity, command success, shared access, and incomplete or overwritten evidence.
8. Appendices. Include labelled screenshots, selected Volatility output, the completed timeline table, and the completed evidence log.

Timeline template

Use a completed version of this table in your report or evidence appendix. Add rows for all material events and keep the main timeline in UTC.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Browser or process evidence					
Download-folder review					
Removable-device evidence					
Drive mapping					
Command-history evidence					
Supported conclusion					

Evidence log template

Use an evidence log that lets an assessor locate the exact Volatility plugin, command, output, registry path, process record, or screenshot supporting each finding.

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E01				
E02				
E03				
E04				
E05				
E06				